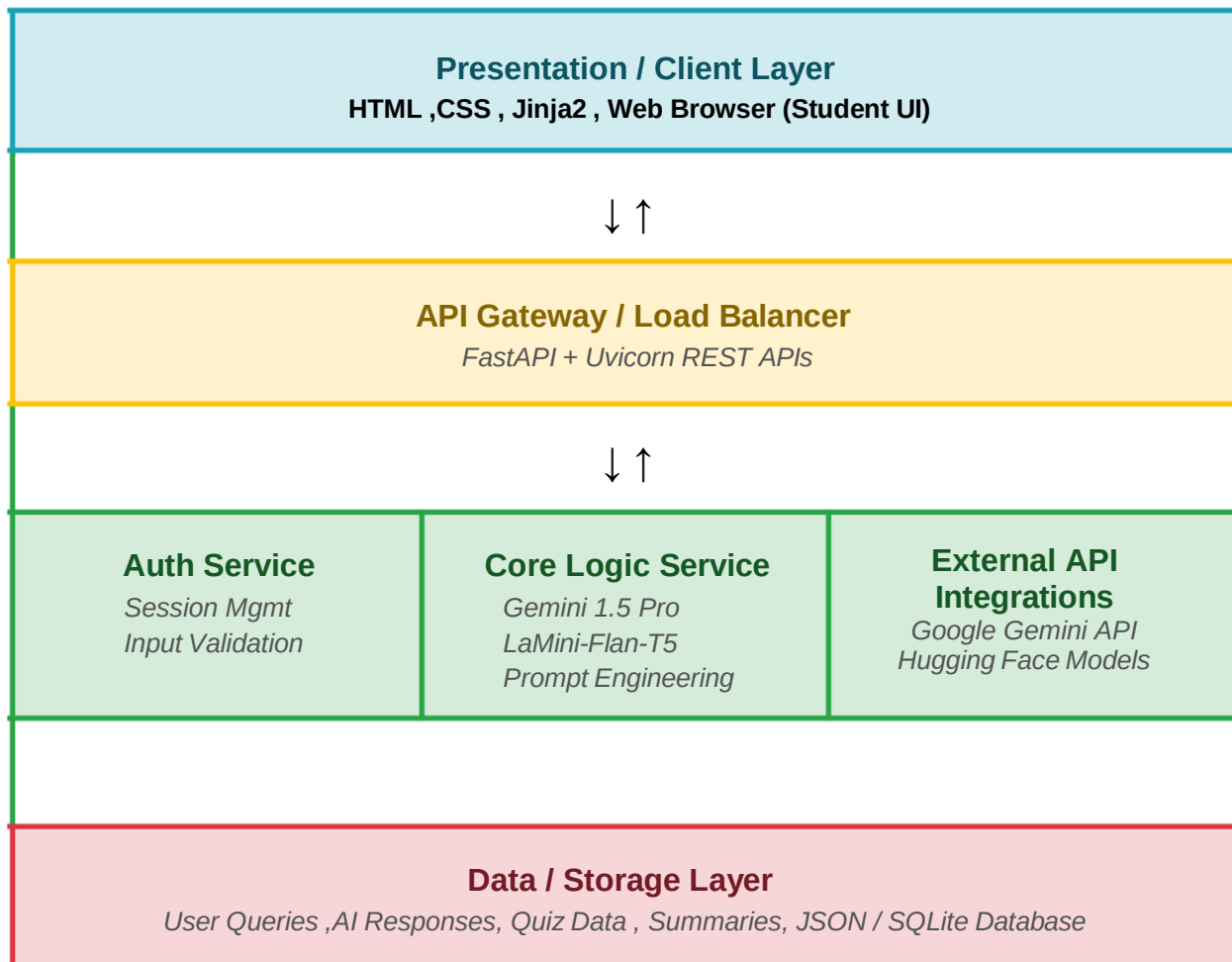


Solution Architecture

Date	30 th June 2026
Team ID	XXXXXXX
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where students can submit questions, generate quizzes, read summaries, and receive learning recommendations.	HTML5, CSS3, Jinja2
API Gateway	Receives HTTP requests from users, routes them to backend services, validates requests, and returns AI-generated responses.	FastAPI, Uvicorn
Authentication Service	Manages user sessions and validates requests before accessing AI services.	FastAPI Security, Session Management
Core Logic Service	Processes educational queries, performs prompt engineering, invokes AI models, formats responses, and handles quiz generation, summarization, explanations, and recommendations.	Python, Google Gemini 1.5 Pro, LaMini-Flan-T5, Transformers, Torch
External API Integrations	Connects to cloud AI services for advanced language generation and educational assistance.	Google Generative AI SDK, Hugging Face Transformers
Database / Storage Layer	Stores user queries, generated quizzes, summaries, AI responses, and learning recommendation history.	SQLite (or JSON files), SQLAlchemy